

Measuring the Mobile Gap: From Network Performance to Infrastructure Criticality

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ABSTRACT

Mobile networks are evolving rapidly, with 5G promising transformative gains in speed, reliability, and latency. However, these advancements do not always translate to improved user experience, due to device limitations, congestion, and inconsistent infrastructure deployment. This disconnect between network capability and actual user experience raises key questions about the design, measurement, and investment in mobile broadband systems. My research addresses this gap through the use of edge user measurements of mobile broadband performance. These measurements predominantly consist of crowdsourced network performance measurements and controlled experiments. Broadly, I develop statistical methods to analyze real-world performance variability and also identify performance bottlenecks, and utilize insight from my analyses to inform infrastructure planning and policy, helping ensure that technological advances lead to more consistent and equitable user outcomes. This extended abstract presents results from recent analyses, outlines my methodological approach, and discusses future directions aimed at advancing both technical understanding and policy design in the mobile broadband landscape.

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1 INTRODUCTION

Mobile broadband plays a vital role in modern life, supporting access to education, healthcare, employment, and other important services. With the rollout of 5G in 2019, mobile networks have promised significant improvements in bandwidth, latency, coverage and connectivity. However, despite technological advancements, the performance that users experience often varies widely, and sometimes fails to meet even the baseline expectations for connectivity [5]. This persistent variability has implications not just for understanding mobile network systems, but also for planning infrastructure and evaluating broadband access policies.

While recent research has examined 5G networks, including mmWave deployments, beam management, mobility, control-plane design, and end-user performance [12–16, 20–23, 34–36], there is

significantly less work on how such designs translate into real-world performance variability across devices, locations, and time. Understanding this interaction is particularly important as 5G networks become more mainstream and users encounter them in diverse conditions.

Speed test platforms, such as Ookla Speedtest [4], M-Lab's NDT7 [3], FCC [10], FAST [2], and Cloudflare [1], are widely-used tools for capturing user-side performance. They provide a unique opportunity to observe real-world mobile network behavior in diverse devices, locations, and timescales. Prior work [6, 7, 17, 18, 37], has discussed what these tools measure and how they operate, but gaps remain in how to interpret and contextualize their results, especially in the vast assortment of operational network conditions.

Despite any limitations and/or biases of such data [8, 25], their broad geographic and temporal coverage makes them a valuable complement to in-depth, network performance studies [19, 24, 26, 29–31]. Specifically, speed test data is now used as part of the FCC's mobile coverage challenge process [9] for cellular networks, which enables individuals, organizations, and local governments to dispute carrier-submitted availability claims. This growing role of crowdsourced speed test data highlights the need for principled methods that not only contextualize these measurements accurately but also relate them to underlying technical factors and infrastructure limitations. Utilizing this data, and going beyond isolated performance snapshots, my work aims to develop a broader understanding of how mobile broadband quality varies across space and time, crucial for both technical insight and policy design. By jointly analyzing mobile network performance and its variability, I aim to uncover patterns in network behavior that traditional studies often overlook, and that could potentially inform more equitable infrastructure investment. To that end, my research focuses on two critical questions:

- (1) What network and device features contribute most significantly to the variability of in-the-wild 5G performance?
- (2) How can we use understanding of the interaction between network, device, and performance to guide effective and equitable network infrastructure investments?

2 QUANTIFYING VARIABILITY

Recent measurement studies have highlighted considerable variability in user-experienced cellular network performance. For instance, in my prior work [30, 32], I analyzed nearly 1.75 million crowdsourced Ookla cellular Speedtest measurements collected over three years from eight U.S. I found substantial variability in 5G performance, as shown in Figure 1a; notably, the poorest-performing 36% of 5G measurements recorded download speeds lower than the top 25% of 4G measurements, as shown in Figure 1b. This variability arises under diverse 5G deployment conditions, where multiple

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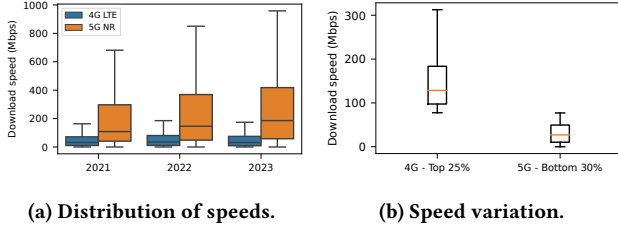


Figure 1: Characterizing 4G and 5G download speed

overlapping factors interact in complex ways, making it difficult to isolate individual effects through direct correlation analysis. To understand the drivers of this variability, I developed a novel statistical approach based on Kullback-Leibler (KL) divergence [11] to compare performance distributions between different network and device configurations. This method helps identify when performance meaningfully diverges between groups and which features co-vary with those differences. I quantified the influence of several key features, including device chipset, frequency band, and physical-layer (PHY) parameters such as signal strength, signal quality, and channel quality index (CQI), as shown in Figure 2. My analysis revealed that while PHY layer conditions contribute to performance differences, they do not fully account for the wide variability observed in 5G networks. Device capabilities (hardware and software), and in particular support for higher-frequency bands, play a key role in determining whether users can fully benefit from available 5G capability. Infrastructure characteristics such as tower density become more relevant at higher frequencies due to smaller wavelengths and interference, but device limitations remain the primary constraint. My controlled experiments further confirmed that older devices failed to connect to available higher-frequency bands, even under identical location and network conditions, reinforcing the impact of hardware architecture on user experience.

3 INFORMING EQUITABLE INFRASTRUCTURE AND POLICY

To complement my analysis of user-side performance variability, I investigate how infrastructure measurements can inform policy outcomes in the mobile broadband landscape, where availability data is typically coarse, self-reported, and rarely validated. In light of these limitations, my work [31] has utilized empirical, crowd-sourced mobile network performance measurements, specifically nationwide Ookla Speedtest data and cell deployment records from 2021 to 2023, to understand how cellular infrastructure and performance have evolved. This study revealed significant disparities in mobile infrastructure and performance. For instance, urban areas had up to 40 times more cellular infrastructure than rural areas, and the improvement in download speeds varied widely between states, with some regions showing little to no gains. These findings raised important questions about the uneven progress of mobile connectivity and its implications for digital equity. To understand how mobile networks complement fixed broadband, I analyzed nationwide coverage and performance at the census block level. While 90% of blocks meet the 25/3 Mbps benchmark, gigabit access is available in only 57% of urban, 38% of rural, and 48% of tribal

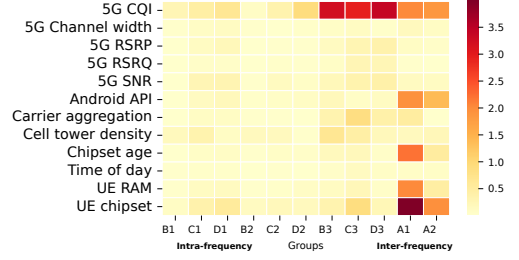


Figure 2: KL Divergence in throughput for a carrier

areas. I also found that 118,000 populated blocks lack any fixed broadband service. In under-served urban blocks, mobile tower density is 1.8 times higher than in well-served ones, suggesting that mobile infrastructure is often deployed to fill fixed coverage gaps. In these same areas, Fixed Wireless Access (FWA) delivers 22% higher speeds in urban regions and 85% higher in rural regions when it does not coexist with wireline broadband.

These findings demonstrate the complex interaction between mobile and fixed broadband systems, which cannot be ignored when planning equitable infrastructure investments.

4 CURRENT WORK

A key insight from recent measurement efforts, including my own, is that mobile broadband performance is variable, and the variability is shaped by a complex interplay of factors ranging from device hardware and spectrum band to signal quality and congestion. While these drivers are well-known, isolating the individual impact of each of these factors is infeasible to do at scale. Rather than attributing performance to specific factors, I focus on modeling the joint behavior of performance and its variability across space and time. This abstraction provides a compact and interpretable representation of mobile broadband quality, capturing not only how mobile networks perform, but also how consistently they do so across different contexts. Hence, my current research focuses on developing robust statistical models that capture how end-user performance varies across space and time. Extending prior modeling studies [27, 28, 33], this work will integrate large-scale crowd-sourced data with measurements from controlled experiments to accurately map variations in network throughput, latency, and signal metrics, evaluating both performance and its consistency across space and time.

To do so, I model mobile broadband performance as a time series and characterize how performance and its variability behave across different locations over time. To account for the well-known data sparsity and uneven measurement density in crowdsourced platforms, I explore interpolation techniques that preserve spatial and temporal coherence and improve the reliability and comparability of the model estimates.

These more complete representations of performance and variability then allow for distinguishing between consistent performance gaps driven by infrastructure limitations and temporary fluctuations caused by local network conditions, thereby informing infrastructure and policy decisions for improved and equitable mobile broadband connectivity.

REFERENCES

- [1] 2023. Cloudflare Speed Test. <https://speed.cloudflare.com>
- [2] 2023. Fast.com. <https://fast.com>
- [3] 2023. Measurement Lab Speed Test. <https://speed.measurementlab.net/#/>
- [4] 2023. Ookla's Speedtest.net. <https://www.speedtest.net>
- [5] Business Insider. 2025. The title of the Business Insider article is: "5G internet was supposed to change the world. Here's why it fell flat." <https://www.businessinsider.com/why-5g-failed-to-change-the-world-2025-3>
- [6] Igor Canadi, Paul Barford, and Joel Sommers. 2012. Revisiting Broadband Performance. In IMC.
- [7] David D. Clark and Sara Wedeman. 2021. Measurement, Meaning and Purpose: Exploring the M-Lab NDT Dataset. In TPRC.
- [8] Nick Feamster and Jason Livingood. 2020. Measuring Internet Speed: Current Challenges and Future Recommendations. *Communications of the ACM* (2020).
- [9] Federal Communications Commission. 2021. Mobile Availability Challenge Resources. <https://help.bdc.fcc.gov/hc/en-us/articles/8779977939227-Mobile-Availability-Challenge-Resources#:~:text=Choose%20the%20Challenge%20Speed%20Test,to%20the%20provider%20for%20response>.
- [10] Federal Communications Commission. 2023. How to Use the FCC Speed Test App to Challenge Mobile Coverage. <https://help.bdc.fcc.gov/hc/en-us/articles/10468786141723-How-to-Use-the-FCC-Speed-Test-App-to-Challenge-Mobile-Coverage>
- [11] Lin Feng, Huibing Wang, Bo Jin, Hao hao Li, Mingliang Xue, and Le Wang. 2019. Learning a Distance Metric by Balancing KL-Divergence for Imbalanced Datasets. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* (2019).
- [12] Rostand AK Fezeu, Eman Ramadan, Wei Ye, Benjamin Minneci, Jack Xie, Arvind Narayanan, Ahmad Hassan, Feng Qian, Zhi-Li Zhang, Jaideep Chandrashekar, et al. 2023. An In-Depth measurement analysis of 5G mmwave PHY latency and its impact on End-to-End delay. In PAM.
- [13] Moinak Ghoshal, Imran Khan, Z. Jonny Kong, Phuc Dinh, Jiayi Meng, Y. Charlie Hu, and Dimitrios Koutsonikolas. 2023. Performance of Cellular Networks on the Wheels. In IMC.
- [14] Ahmad Hassan, Arvind Narayanan, Anlan Zhang, Wei Ye, Ruiyang Zhu, Shuowei Jin, Jason Carpenter, Z. Morley Mao, Feng Qian, and Zhi-Li Zhang. 2022. Vivisectioning Mobility Management in 5G Cellular Networks. In *Proceedings of ACM SIGCOMM '22*.
- [15] Rostand A. K. Fezeu, Claudio Fiandrino, Eman Ramadan, Jason Carpenter, Lilian Coelho de Freitas, Faaiq Bilal, Wei Ye, Joerg Widmer, Feng Qian, and Zhi-Li Zhang. 2024. Unveiling the 5G Mid-Band Landscape: From Network Deployment to Performance and Application QoE. In SIGCOMM (Sydney, NSW, Australia).
- [16] Yanbing Liu and Chunyi Peng. 2023. A Close Look at 5G in the Wild: Unrealized Potentials and Implications. In *IEEE INFOCOM 2023 - IEEE Conference on Computer Communications*.
- [17] Kyle MacMillan, Tarun Mangla, Marc Richardson, and Nick Feamster. 2022. Best Practices for Collecting Speed Test Data. In TPRC.
- [18] Kyle MacMillan, Tarun Mangla, James Saxon, Nicole P. Marwell, and Nick Feamster. 2023. A Comparative Analysis of Ookla Speedtest and Measurement Labs Network Diagnostic Test (NDT7). In *Proceedings ACM SIGMETRICS '23*.
- [19] Tarun Mangla, Udit Paul, Arpit Gupta, Nicole P. Marwell, and Nick Feamster. 2022. Internet Inequity in Chicago: Adoption, Affordability, and Availability. (August 2022). Available at SSRN: <https://ssrn.com/abstract=4182994>.
- [20] Arvind Narayanan, Eman Ramadan, Jason Carpenter, Qingxu Liu, Yu Liu, Feng Qian, and Zhi-Li Zhang. 2020. A First Look at Commercial 5G Performance on Smartphones. In WWW.
- [21] Arvind Narayanan, Eman Ramadan, Rishabh Mehta, Xinyue Hu, Qingxu Liu, Rostand A. K. Fezeu, Udhaya Kumar Dayalan, Saurabh Verma, Peiqi Ji, Tao Li, Feng Qian, and Zhi-Li Zhang. 2020. Lumos5G: Mapping and Predicting Commercial MmWave 5G Throughput. In IMC.
- [22] Arvind Narayanan, Muhammad Iqbal Rochman, Ahmad Hassan, Bariq S. Firmansyah, Vanlin Sathya, Monisha Ghosh, Feng Qian, and Zhi-Li Zhang. 2022. A Comparative Measurement Study of Commercial 5G mmWave Deployments. In *IEEE INFOCOM*.
- [23] Arvind Narayanan, Xumiao Zhang, Ruiyang Zhu, Ahmad Hassan, Shuowei Jin, Xiao Zhu, Xiaoxuan Zhang, Denis Rybkin, Zhengxuan Yang, Zhuoqing Morley Mao, Feng Qian, and Zhi-Li Zhang. 2021. A Variegated Look at 5G in the Wild: Performance, Power, and QoE Implications. In *Proceedings of ACM SIGCOMM '21*.
- [24] Udit Paul, Jiamo Liu, Vivek Adarsh, Mengyang Gu, Arpit Gupta, and Elizabeth Belding. 2021. Characterizing Performance Inequity Across U.S. Ookla Speedtest Users. <https://arxiv.org/abs/2110.12038>.
- [25] Udit Paul, Jiamo Liu, Mengyang Gu, Arpit Gupta, and Elizabeth Belding. 2022. The Importance of Contextualization of Crowdsourced Active Speed Test Measurements. In IMC.
- [26] Reinaldo Sanchez-Arias, Luis G. Jaimes, Shahram Taj, and Md. Selim Habib. 2023. Understanding the State of Broadband Connectivity: An Analysis of Speedtests and Emerging Technologies. *IEEE Access* (2023).
- [27] Taveesh Sharma, Jonatas Marques, Nick Feamster, and Nicole P. Marwell. 2023. A First Look at the Spatial and Temporal Variability of Internet Performance Data in Hyperlocal Geographies. Available at SSRN: <https://ssrn.com/abstract=4182994>.
- [28] Taveesh Sharma, Paul Schmitt, Francesco Bronzino, Nick Feamster, and Nicole P. Marwell. 2024. Beyond Data Points: Regionalizing Crowdsourced Latency Measurements. (2024).
- [29] Joel Sommers and Paul Barford. 2012. Cell vs. WiFi: On the Performance of Metro Area Mobile Connections. In *Proceedings of the Internet Measurement Conference (IMC '12)*.
- [30] Liu Jiayi Srinivasavaradhan, Varshika and Elizabeth Belding. 2025. 5G Performance: A Multidimensional Variability Analysis. In PAM.
- [31] Varshika Srinivasavaradhan, Owen Park, and Elizabeth Belding. 2024. Mapping Cellular Network Evolution and Infrastructure Criticality: A Nationwide Analysis. In TPRC.
- [32] Varshika Srinivasavaradhan, Udit Paul, Ellen Zegura, and Elizabeth M. Belding. 2024. Poster: Beyond the Bars: Decoding the Complexities of Cellular Network Performance through Speedtest Data. In *Proceedings of the 25th International Workshop on Mobile Computing Systems and Applications (HotMobile '24)*.
- [33] Jing Wang, Jian Tang, Zhiyuan Xu, Yanzhi Wang, Guoliang Xue, Xing Zhang, and Dejun Yang. 2017. Spatiotemporal modeling and prediction in cellular networks: A big data enabled deep learning approach. In *IEEE INFOCOM 2017 - IEEE Conference on Computer Communications*.
- [34] Dongzhu Xu, Anfu Zhou, Xinyu Zhang, Guixian Wang, Xi Liu, Congkai An, Yiming Shi, Liang Liu, and Huadong Ma. 2020. Understanding Operational 5G: A First Measurement Study on Its Coverage, Performance and Energy Consumption. In *Proceedings of ACM SIGCOMM '20*.
- [35] Wei Ye, Xinyue Hu, Steven Sleder, Anlan Zhang, Udhaya Kumar Dayalan, Ahmad Hassan, Rostand A. K. Fezeu, Akshay Jajoo, Myungjin Lee, Eman Ramadan, Feng Qian, and Zhi-Li Zhang. 2024. Dissecting Carrier Aggregation in 5G Networks: Measurement, QoE Implications and Prediction. In SIGCOMM (Sydney, NSW, Australia).
- [36] Xinjie Yuan, Mingzhou Wu, Zhi Wang, Yifei Zhu, Ming Ma, Junjian Guo, Zhi-Li Zhang, and Wenwu Zhu. 2022. Understanding 5G Performance for Real-World Services: A Content Provider's Perspective. In *Proceedings of ACM SIGCOMM '22*.
- [37] Zesen Zhang, Jiting Shen, and Ricky K. P. Mok. 2024. Empirical Characterization of Ookla's Speed Test Platform: Analyzing Server Deployment, Policy Impact, and User Coverage. In *CCWC*.